

Linear Combinations

If $\vec{w}$ is the vector in vector space V , then $\vec{w}$ is said to be a linear combination of the vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r$ in V , if $\vec{w}$ can be expressed in the form $\vec{w} = k_1\vec{v}_1 + k_2\vec{v}_2 + \dots + k_r\vec{v}_r$, where $k_1, k_2, \dots, k_r$ are scalars. These scalars are called the coefficients of linear combinations.

THEOREM 4.2.3

If S is a nonempty set of vectors in a vector space V , then:

(a) The set W of all possible linear combinations of the vectors in S is a subspace of V .

(b) The set W in part (a) is the “smallest” subspace of V that contains all of the vectors in S in the sense that any other subspace that contains those vectors contains W .

Example 1:

Let $\vec{v}_1 = (1, 0, 0)$, $\vec{v}_2 = (0, 1, 0)$, $\vec{v}_3 = (0, 0, 1)$ in R^3 . Then $\vec{w} = (2, 3, 4)$ can be written as linear combinations of $\vec{v}_1, \vec{v}_2, \vec{v}_3$.

Solution:

$$\vec{w} = k_1\vec{v}_1 + k_2\vec{v}_2 + k_3\vec{v}_3$$

$$(2, 3, 4) = k_1(1, 0, 0) + k_2(0, 1, 0) + k_3(0, 0, 1)$$

$$(2, 3, 4) = 2(1, 0, 0) + 3(0, 1, 0) + 4(0, 0, 1)$$

$$\vec{w} = 2\vec{v}_1 + 3\vec{v}_2 + 4\vec{v}_3$$

Example 2:

Consider the vectors $\vec{u} = (1, 2, -1)$ and $\vec{v} = (6, 4, 2)$ in R^3 . Show that $\vec{w} = (9, 2, 7)$ is linear combination of $\vec{u}$ and $\vec{v}$.

Also prove that $\vec{w}' = (4, -1, 8)$ is not linear combination of $\vec{u}$ and $\vec{v}$.

Solution:

In order to form linear combination of $\vec{u}$ and $\vec{v}$, there exist k_1 and k_2 such that

$$\vec{w} = k_1\vec{u} + k_2\vec{v}$$

$$(9, 2, 7) = k_1(1, 2, -1) + k_2(6, 4, 2)$$

$$(9, 2, 7) = (k_1, 2k_1, -k_1) + (6k_2, 4k_2, 2k_2)$$

$$(9, 2, 7) = (k_1 + 6k_2, 2k_1 + 4k_2, -k_1 + 2k_2)$$

Equating corresponding components gives:

$$k_1 + 6k_2 = 9$$

$$2k_1 + 4k_2 = 2$$

$$-k_1 + 2k_2 = 7$$

Solving this system by using Gauss Elimination:

$$\left[\begin{array}{cc|c} 1 & 6 & 9 \\ 2 & 4 & 2 \\ -1 & 2 & 7 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1 & 6 & 9 \\ 0 & -8 & -16 \\ 0 & 8 & 16 \end{array} \right] \text{ } R_2 - 2R_1 \text{ and } R_3 + R_1$$

$$\left[\begin{array}{cc|c} 1 & 6 & 9 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{array} \right] \text{ } -\frac{R_2}{8} \text{ and } \frac{R_3}{8}$$

$$\left[\begin{array}{cc|c} 1 & 6 & 9 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right] \text{ } R_3 - R_2$$

$$\begin{cases} k_1 + 6k_2 = 9 & \text{--- (1)} \\ k_2 = 2 \\ 0 = 0 \end{cases}$$

Put value of $k_2 = 2$ in (1), we get:

$$k_1 = -3$$

So

$$\vec{w} = -3\vec{u} + 2\vec{v}$$

That is

$$(9, 2, 7) = -3(1, 2, -1) + 2(6, 4, 2)$$

So $\vec{w}$ is the linear combination of $\vec{u}$ and $\vec{v}$.

Similarly, we have to check whether $\vec{w}'$ is linear combination of $\vec{u}$ and $\vec{v}$.

If $\vec{w}'$ is linear combination of $\vec{u}$ and $\vec{v}$, then there must exist k_1 and k_2 , such that:

$$\vec{w'} = k_1 \vec{u} + k_2 \vec{v}$$

$$(4, -1, 8) = k_1(2, 2, -1) + k_2(6, 4, 2)$$

$$(4, -1, 8) = (2k_1 + 6k_2, 2k_1 + 4k_2, -k_1 + 2k_2)$$

$$\begin{cases} 4 = k_1 + 6k_2 \\ -1 = 2k_1 + 4k_2 \\ 8 = -k_1 + 2k_2 \end{cases}$$

$$\begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 6 & 4 \\ 0 & -8 & -9 \\ 0 & 8 & 12 \end{bmatrix} \quad R_2 - 2R_1 \quad \text{and} \quad R_3 + R_1$$

$$\begin{bmatrix} 1 & 6 & 4 \\ 0 & 1 & \frac{9}{8} \\ 0 & 2 & 3 \end{bmatrix} \quad -\frac{1}{8}R_2 \quad \text{and} \quad \frac{1}{4}R_3$$

$$\begin{bmatrix} 1 & 6 & 4 \\ 0 & 1 & \frac{9}{8} \\ 0 & 0 & \frac{3}{2} \end{bmatrix} \quad R_3 - 2R_2$$

This system has no solution. So, no such k_1 and k_2 exist. So $\vec{w'}$ is not the linear combination of $\vec{u}$ and $\vec{v}$.

Example 3:

Which of the following are linear combination of $\vec{u} = (0, -2, 2)$ and $\vec{v} = (1, 3, -1)$.

- a) (2, 2, 2)
- b) (3, 1, 5)
- c) (0, 4, 5)
- d) (0, 0, 0)

Example 4:

Express the following as linear combination of $\vec{u} = (2,1,4)$, $\vec{v} = (1,-1,3)$, $\vec{w} = (3,2,5)$

- a) $(-9, -7, -15)$
- b) $(66, 11, 6)$
- c) $(0,0,0)$
- d) $(7,8,9)$

Example 5: Which of the following is the linear combination of

$$A = \begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$$

- a) $M = \begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix}$
- b) $N = \begin{bmatrix} 6 & 0 \\ 3 & 8 \end{bmatrix}$
- c) $O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
- d) $P = \begin{bmatrix} -1 & 5 \\ 7 & 1 \end{bmatrix}$

Solution: In order to form linear combination of A, B and C , we write

$$M = k_1 A + k_2 B + k_3 C$$

$$\begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix} = k_1 \begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix} + k_2 \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} + k_3 \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix} = \begin{bmatrix} 4k_1 & 0 \\ -2k_1 & -2k_1 \end{bmatrix} + \begin{bmatrix} k_2 & -k_2 \\ 2k_2 & 3k_2 \end{bmatrix} + \begin{bmatrix} 0 & 2k_3 \\ k_2 & 4k_3 \end{bmatrix}$$

$$\begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix} = \begin{bmatrix} 4k_1 + k_2 & -k_2 + 2k_3 \\ -2k_1 + 2k_2 + k_3 & -2k_1 + 3k_2 + 4k_3 \end{bmatrix}$$

$$6 = 4k_1 + k_2 \quad \text{--- (1)}$$

$$-8 = -k_2 + 2k_3 \quad \text{--- (2)}$$

$$-1 = -2k_1 + 2k_2 + k_3 \quad \text{--- (3)}$$

$$-8 = -2k_1 + 3k_2 + 4k_3 \quad \text{--- (4)}$$

Add (1) and (2):

$$-2 = 4k_1 + 2k_3 \quad \text{or} \quad -1 = 2k_1 + k_3$$

$$k_3 = -1 - 2k_1, \text{ put in (3)}$$

$$-1 = -2k_1 + 2k_2 + k_3$$

$$-1 = -2k_1 + 2k_2 + (-1 - 2k_1)$$

$$-1 = -2k_1 + 2k_2 - 1 - 2k_1$$

$$-4k_1 + 2k_2 = 0 \quad \text{--- (5)}$$

Add (1) and (5)

$$3k_2 = 6$$

$$k_2 = 2, \text{ Put in (1)}$$

$$4k_1 + 2 = 6$$

$$k_1 = 1$$

$k = 2$, put in (2)

$$-8 = -2 + 2k_3$$

$$-8 + 2 = 2k_3$$

$$k_3 = -3$$

Put $k_1 = 1, k_2 = 2, k_3 = -3$ in (4)

$$-8 = -2(1) + 3(2) + 4(-3)$$

$$-8 = -2 + 6 - 12$$

$$-8 = -8$$

$$M = 1A + 2B - 3C$$

$$\begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix} + 2 \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} - 3 \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$$

So $\begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix}$ is linear combination of A, B and C.

Example 6: In each part the vectors are as linear combination of $P_1 = 2 + x + 4x^2$, $P_2 = 1 - x + 3x^2$ and $P_3 = 3 + 2x + 5x^2$

- a) $-9 - 7x - 15x^2$
- b) $6 + 11x + 6x^2$
- c) 0
- d) $7 + 8x + 9x^2$

Solution: In order to form linear combination, we write

$$-9 - 7x - 15x^2 = k_1P_1 + k_2P_2 + k_3P_3$$

$$-9 - 7x - 15x^2 = k_1(2 + x + 4x^2) + k_2(1 - x + 3x^2) + k_3(3 + 2x - 5x^2)$$

$$= (2k_1 + k_2 + 3k_3) + (k_1x - k_2x + 2k_3x) + (4k_1x^2 + 3k_2x^2 - 5k_3x^2)$$

$$-9 - 7x - 15x^2 = (2k_1 + k_2 + 3k_3) + (k_1 - k_2 + 2k_3)x + (4k_1 + 3k_2 - 5k_3)x^2$$

Comparing equations on both sides, we get:

$$\begin{cases} -9 = 2k_1 + k_2 + 3k_3 & \text{--- (1)} \\ -7 = k_1 - k_2 + 2k_3 & \text{--- (2)} \\ -15 = 4k_1 + 3k_2 - 5k_3 & \text{--- (3)} \end{cases}$$

$$\begin{bmatrix} 2 & 1 & 3 & -9 \\ 1 & -1 & 2 & -7 \\ 4 & 3 & -5 & -15 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 2 & -7 \\ 2 & 1 & 3 & -9 \\ 4 & 3 & -5 & -15 \end{bmatrix} \quad R_{12}$$

$$\begin{bmatrix} 1 & -1 & 2 & -7 \\ 0 & 3 & -1 & 5 \\ 0 & 7 & -13 & 13 \end{bmatrix} \quad R_2 - 2R_1, \quad R_3 - 4R_1$$

$$\begin{bmatrix} 1 & -1 & 2 & -7 \\ 0 & 3 & -1 & 5 \\ 0 & 1 & -11 & 3 \end{bmatrix} \quad R_3 - 2R_2$$

$$\begin{bmatrix} 1 & -1 & 2 & -7 \\ 0 & 1 & -11 & 3 \\ 0 & 3 & -1 & 5 \end{bmatrix} \quad R_{23}$$

$$\begin{bmatrix} 1 & 0 & -9 & -4 \\ 0 & 1 & -11 & 3 \\ 0 & 0 & 32 & -4 \end{bmatrix} \quad R_1 + R_2, \quad R_3 - 3R_2$$

$$\begin{bmatrix} 1 & 0 & -9 & -4 \\ 0 & 1 & -11 & 3 \\ 0 & 0 & 1 & -\frac{1}{8} \end{bmatrix} \quad \frac{1}{32}R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{41}{8} \\ 0 & 1 & 0 & \frac{13}{8} \\ 0 & 0 & 1 & -\frac{1}{8} \end{bmatrix} \quad R_1 + 9R_3, R_2 + 11R_3$$

As $k_1 = -\frac{41}{8}, k_2 = \frac{13}{8}, k_3 = -\frac{1}{8}$

$$-9 - 7x - 15x^2 = -\frac{41}{8}(2 + x + 4x^2) + \frac{13}{8}(1 - x + 3x^2) - \frac{1}{8}(3 + 2x - 5x^2)$$

So $-9 - 7x - 15x^2$ is linear combination of P_1, P_2, P_3 .